

VARSHIKA S

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LeetCode: https://leetcode.com/u/Varshika_Saravanan/

EDUCATION

Sri Ramakrishna Engineering College, Coimbatore

B.Tech Information Technology (2023–2027) | CGPA: 9.06/10

TECHNICAL SKILLS

Programming: Python, Java, JavaScript, SQL

AI/ML: Machine Learning, Generative AI, LangChain, Ollama, Qwen 2.5, OpenCV, NumPy, Pandas

Databases: MongoDB, MySQL

Web Development: Flask, React.js, HTML, CSS

Tools: Git, GitHub, Google Colab, AWS, Snowflake, MATLAB, Simulink

Concepts: RAG, Vector Databases, Semantic Search, Prompt Engineering, REST APIs, JSON

PROJECTS

AI Learning Companion – Local RAG Educational Assistant

- Developed a fully local Retrieval-Augmented Generation (RAG) educational assistant using LangChain, FAISS, Ollama, and Qwen 2.5.
- Implemented document ingestion, chunking, embedding generation, semantic retrieval, and context-aware question answering.
- Built summarization, flashcard generation, quiz creation, concept extraction, and personalized study planning features.

Water Footprint Calculator

- Developed a machine learning-based Flask application to estimate water consumption using crop, location, and cultivation data.
- Implemented data preprocessing, prediction workflows, and visualization modules.
- Added historical tracking and reporting features for data-driven decision-making.

SDV Lane Keeping Assistant

- Developed a lane detection system using OpenCV and computer vision techniques.

- Implemented thresholding, perspective transformation, and lane tracking algorithms.
- Optimized image processing pipelines for reliable performance.

INTERNSHIP

Cyber Security Intern | Hackup Technology | Jun 2025 – Jul 2025

- Analyzed security vulnerabilities and attack patterns using industry-standard tools.
- Worked on network monitoring, Linux environments, SQL Injection, and XSS concepts.
- Gained practical exposure to cybersecurity workflows and troubleshooting techniques.

CERTIFICATIONS & ACHIEVEMENTS

- NPTEL Artificial Intelligence – Top 1% (Score: 92%)
- AWS Cloud Foundations
- SnowPro Associate Platform
- NIELIT Robotic Process Automation Bootcamp
- MATLAB Onramp
- Simulink Onramp
- Published Research Paper in IJARCS